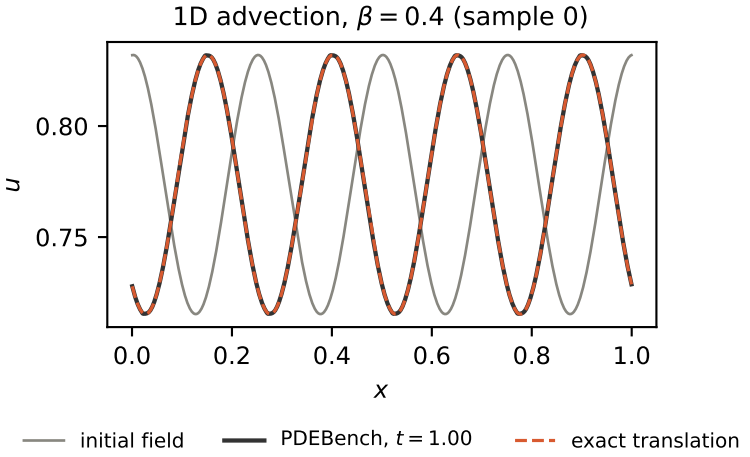
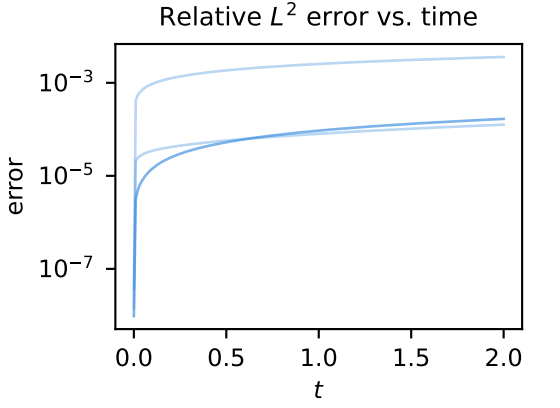
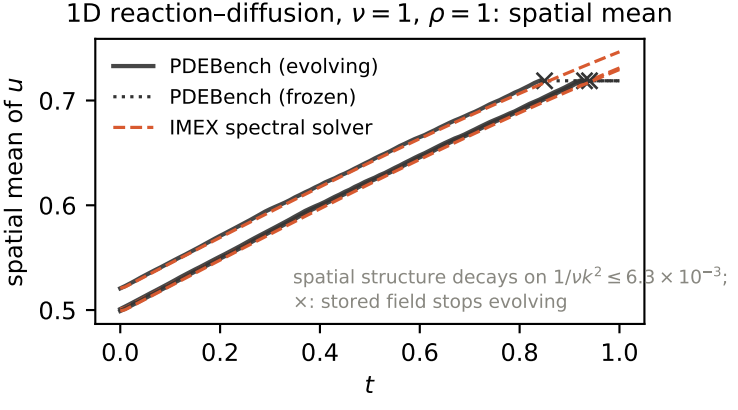
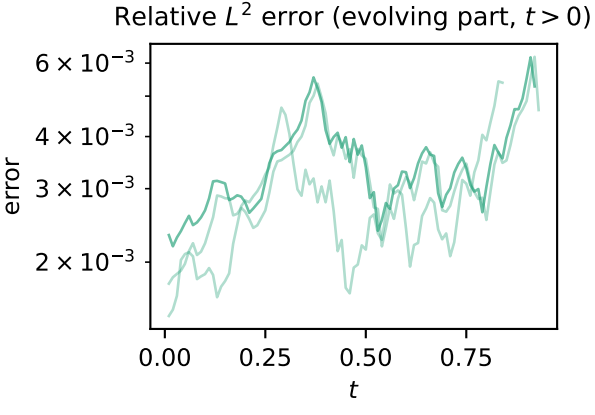

1D advection, $\beta = 0.4$ (sample 0)
0.80
0.75
u
0.0
0.2
0.4
0.6
0.8
1.0
x
initial field
PDEBench, $t = 1.00$
exact translation


Relative L^2 error vs. time
10^{-3}
10^{-5}
10^{-7}
error
0.0
0.5
1.0
1.5
2.0
t


1D reaction–diffusion, $\nu = 1$, $\rho = 1$: spatial mean
PDEBench (evolving)
PDEBench (frozen)
IMEX spectral solver
spatial mean of u
0.7
0.6
0.5
spatial structure decays on $1/\nu k^2 \leq 6.3 \times 10^{-3}$;
×: stored field stops evolving
0.0
0.2
0.4
0.6
0.8
1.0
t


Relative L^2 error (evolving part, $t > 0$)
6×10^{-3}
4×10^{-3}
3×10^{-3}
2×10^{-3}
error
0.00
0.25
0.50
0.75
t